AM RADIO PLAYER PROJECT JOURNAL

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Special thanks to my dad for giving me helpful advice during the project.

Date: 09/05/26

Current Project Stage: (1) Problem clarification.

Current Problem: What goes into designing an AM radio?

Research Tasks:

- ~~1. Learn about AM/FM general info.~~
- ~~2. Identify radio components to be built.~~
- 3. Do research on antenna design
- 4. Do research on LC tuners
- 5. Do research on signal amplifiers
- 6. Do research on DSP AM demodulation techniques
- 7. Do research on speakers

Deadline: 11:59 PM, 16/05/26.

Design Tasks:

- 1. Brainstorm case design.
- 2. Investigate a "Diode" crystal radio solution

Deadline: 11:59 PM, 18/05/26.

Comments:

Today, I read up to chapter 3 in the textbook: "MICROWAVE AND RF DESIGN" by Michael Steer, and I have learned about what AM/FM radiowaves are, and how they are mathematically described. I am now also aware of the challenges that go into the detection, filtering, amplification, and demodulation of an AM/FM signal. I have also explored online to find potential solutions to building the radio, and I have found a diode-replacement version of a crystal radio to be quite promising. Investigating other radio solutions is advised.

Goal for next check-in: Complete Research Tasks 3-5, and work on Design Task 1.

Date: 18/05/26

Current Project Stage: (2) Brainstorming Ideas.

Current Problem: How do we put the components of a radio together?

Research Tasks:

- ~~1. Learn about AM/FM general info.~~
- ~~2. Identify radio components to be built.~~
- 3. ~~Do research on antenna design~~
- 4. ~~Do research on LC tuners~~
- 5. ~~Do research on signal amplifiers~~
- 6. ~~Do research on DSP AM demodulation techniques~~
- 7. ~~Do research on speakers~~

Deadline: 11:59 PM, 31/05/26.

Design Tasks:

- 1. Brainstorm case design.
- 2. ~~Investigate a "Diode" crystal radio solution~~
- 3. Finish first radio design schematic
- 4. Construct a test "wire antenna".

Deadline: 11:59 PM, 31/05/26.

Comments:

Today, I did research on the construction of a diode crystal radio, and how theyre amplified. Additionally, I have finished a rough 1st design for a radio with the diode crystal radio setup. Revision is needed for this design.

Goal for next check-in: Dismiss D.T 1, finish D.T 3, start on D.T 4.

Date: 30/05/26

Current Project Stage: (2) Brainstorming Ideas.

Current Problem: How do we put the components of a radio together?

Research Tasks:

- 1.

Deadline:

Design Tasks:

- ~~3. Finish first radio design schematic~~
- 4. Construct a test “wire antenna”.
    - 4.2. Construct first prototype
- 5. Purchase needed materials for first prototype

Deadline: 11:59 PM, 07/06/26.

Comments:

Today, I finished the first design for the radio circuit, and have chosen appropriate component values. Revise design after next check in.

Goal for next check-in: Complete D.T 5.

Date: 01/06/26

Current Project Stage: (3) Idea development/refinement.

Current Problem: How can the design be further improved?

Research Tasks:

- 1.

Deadline:

Design Tasks:

- 4. Construct a test "wire antenna".
    - 4.2. Construct first prototype
    - 4.3. Test first prototype with different C and R values.
- ~~5. Purchase needed materials for first prototype~~

Deadline: 11:59 PM, 07/06/26.

Comments:

Today, I finished most of the first breadboard prototype of the radio, with the exception of the earpiece. Next check in, I will be soldering the earpiece receiver jack to be "breaboardable." Revise design after breadboard prototype + testing are done.

Goal for next check-in: Complete D.T 5.

Date: 13/06/26

Current Project Stage: (3) Idea development/refinement.

Current Problem: How can the design be further improved?

Research Tasks:

- 1.

Deadline:

Design Tasks:

- 4. Construct a test “wire antenna”.
    - 4.2. Construct first prototype
    - 4.3. Test first prototype with different C and R values.
- 5. Purchase needed materials for first prototype (again)
- 6. Try an earth ground connection
- 7. Try the PVC inductor.

Project Deadline: 11:59 PM, 01/07/26.

Comments:

After debugging the radio circuit trying to figure out what was wrong, I was able to conclude that the point of issue was that I had used an incorrect inductor, as well as an incorrect “ground”. Instead, I should have created the PVC self made inductor that I saw when doing my research, as well as using a proper “earth ground” connection instead of a breadboard ground connection.

Goal for next check-in: Complete D.T 5.

Date: 18/06/26

Current Project Stage: (3) Idea development/refinement.

Current Problem: How can the design be further improved?

Research Tasks:

- 1.

Deadline:

Design Tasks:

- 4. Construct a test "wire antenna".
  - ~~4.2. Construct first prototype~~
  - ~~4.3. Test first prototype with different C and R values.~~
  - 4.4. Buy and implement antenna inductor.
- ~~6. Try an earth ground connection~~
- ~~7. Try the PVC inductor.~~

Project Deadline: 11:59 PM, 01/07/26.

Comments:

Today, I implemented an aluminum sheet "ground" for my radio, adjusted my previously incorrect low pass filter to the now correct one (now with a cutoff freq of ~1530 Hz), installed an extra 2 transistor amplifying stages, and updated the electrical schematic in KiCad. By next check in, I will add the missing inductor that the antenna needs to "cancel out" the capacitance of the wire. The most notable thing that happened today was that these fixes had allowed me to hear a recognizable voice through the white noise from the earpiece.

Goal for next check-in: Finalize and complete D.T 4.4.

Date: 20/06/26

Current Project Stage: (3) Idea development/refinement.

Current Problem: How can the design be further improved?

Research Tasks:

- 1.

Deadline:

Design Tasks:

- 4. Construct a test "wire antenna".
  - ~~4.2. Construct first prototype~~
  - ~~4.3. Test first prototype with different C and R values.~~
  - 4.4. Buy and implement antenna inductor.
- ~~6. Try an earth ground connection~~
- ~~7. Try the PVC inductor.~~

Project Deadline: 11:59 PM, 01/07/26.

Comments:

Today, I worked on finding the approximate capacitance of my wire antenna, which I found to be 0.91 pF (rounded to 1 pF to partially account for the real wire having less total electric potential than an infinite wire) , and found the value of an inductor that will work well enough to cancel the impedance of the wire antenna (47 mH, with a +- 10 mH acceptable deviation when purchasing the inductor). I will be purchasing this inductor on the 22nd of June.

Goal for next check-in: Finalize and complete D.T 4.4.

Date: 24/06/26

Current Project Stage: (3) Idea development/refinement.

Current Problem: How can the design be further improved?

Research Tasks:

- 1.

Deadline:

Design Tasks:

- ~~4. Construct a test "wire antenna".~~
  - ~~4.2. Construct first prototype~~
  - ~~4.3. Test first prototype with different C and R values.~~
  - ~~4.4. Buy and implement antenna inductor.~~
- ~~6. Try an earth ground connection~~
- ~~7. Try the PVC inductor.~~
- 8. Improve carrier freq selection
  - 8.2. test cascaded LC tanks
  - 8.3. test wire tap coil solution

Project Deadline: 11:59 PM, 01/07/26.

Comments:

Today, upon closer inspection, the radio was currently playing many different radio stations at the same time. I hypothesize that this is due to poor selectivity of the single LC tank or poor impedance matching between the antenna and the rest of the circuit. To fix this, I will be testing D.T 8.2 and 8.3 to see which solution yields better results.

Goal for next check-in: Complete D.T 8.2 and 8.3.

Date: 28/06/26

Current Project Stage: (3) Idea development/refinement.

Current Problem: How can the design be further improved?

Research Tasks:

- 1.

Deadline:

Design Tasks:

- 8. Improve carrier freq selection
  - 8.2. test cascaded LC tanks
  - 8.3. test wire tap coil solution
- 9. Test an RF amp + 1x audio amp instead of 2x audio amp

Project Deadline: 11:59 PM, 01/07/26.

Comments:

Today, I have worked on designing an LC tuner for my circuit using Ltspice and the LC filter design tool. I Have settled on using a Chebyshev bandpass 2nd order LC filter and have conducted a virtual experiment to verify that the LC tuner will select 730 kHz and significantly attenuate all other signals. I will also need to buy another 10 mH inductor (possibly 2) before completing all D.T 8.x.

Goal for next check-in: Complete D.T 8.2 and 8.3.

Date: 29/06/26

Current Project Stage: (3) Idea development/refinement.

Current Problem: How can the design be further improved?

Research Tasks:

- 1. Figure out why LC tanks are not working.

Deadline:

Design Tasks:

- 8. Improve carrier freq selection
  - ~~8.2. test cascaded LC tanks~~
  - 8.3. test wire tap coil solution
- ~~9. Test an RF amp + 1x audio amp instead of 2x audio amp~~

Project Deadline: 11:59 PM, 07/07/26.

Comments:

Today, I have completed 2/3 D.T left for me, and have seen 0 improvement in carrier frequency selection, despite them theoretically being enough to fix my issues. Unfortunately, the project deadline will need to be extended, as I need to figure out why the LC tanks are not tuning to the proper radio frequency (730kHz). Testing D.T 8.3 is now a top priority.

Goal for next check-in: Complete D.T 8.3 first, and then R.T 1.

Date: 01/07/26

Current Project Stage: (4) Project wrap-up + Future consideration.

Current Problem: What can I do in the future to build on this project?

Research Tasks:

- ~~1. Figure out why LC tanks are not working.~~

Deadline:

Design Tasks:

- ~~8. Improve carrier freq selection~~
  - ~~8.2. test cascaded LC tanks~~
  - ~~8.3. test wire tap coil solution~~
- ~~9. Test an RF amp + 1x audio amp instead of 2x audio amp~~

Project Deadline: 11:59 PM, 01/07/26.

Comments:

Today, I have completed all tasks. By coiling the antenna around one of the tuning inductors, grounding the antenna, and providing a much stronger ground (PC metal casing) I am now able to hear ONLY the 690 AM channel, with no other interfering radio stations. Additionally, the white noise is very quiet too. Since I have found success in tuning to 690 AM, my project goal of listening to 730 AM will be shifted to 690 AM. Since the project has been completed by the original deadline, that is the deadline for the project now. With this, the project is now complete, and optionally, PCB development + production may now be started. Moving forward, I intend to understand further the project in its entirety, as I currently still have some questions about my own design choices and why they worked out.

(05/07/26) Note: extra actions taken by me for this project will still be documented in this journal.

Date: 05/07/26

Current Project Stage: (5) Extra Actions.

Current Problem: What more can be done for this project?

Tasks:

- 1. Order PCB design.

Comments:

Today, I built the PCB design for the radio project. I intend on having a PCB manufacturer make my PCB design. I also intend on designing a basic case for my PCB.

Date: 10/07/26

Current Project Stage: (5) Extra Actions.

Current Problem: What more can be done for this project?

Tasks:

- 1. Order PCB design.
- 2. Order Case

Comments:

Today, I built the case for my radio. I intend on getting tasks 1 and 2 done soon.

Date: 10/07/26

Current Project Stage: (5) Extra Actions.

Current Problem: What more can be done for this project?

Tasks:

- 1. Order PCB design.
- 2. Order Case
- 3. Create product description document

Comments:

Today, I ordered my case to be 3D printed and started working on the product description for the radio.

Date: 18/07/26

Current Project Stage: (6) Project Closure.

Current Problem:

Tasks:

Comments:

Today, I have finalized the product description document, and unfortunately due to unresponsive PCB manufacturers, have cancelled the construction of the final product for the foreseeable future. If the final product is ever constructed, it will not be documented here. Since I have decided that future iterations on this project will become their standalone project(s), there will be no further updates to this document.

Skills learned:

- KiCad schematic and PCB editing
- Ltspice circuit analysis
- Basic LC filter design
- Basic BJT transistor amplification design
- Basic form of impedance matching

Resources consulted:

https://radiowest.ca/vanrad.html
https://www.antenna-theory.com/basics/main.php
https://www.acmaustin.com/lecture_notes/antennas.pdf
https://techlib.com/electronics/crystal.html
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https://www.youtube.com/watch?v=fziUQaVQxA4

https://www.researchgate.net/figure/Relative-magnetic-permeability-of-Mn-Zn-ferrite-vs-frequency-in-the-MF-range-r-r_fig1_327984469

claude code, and chatGPT.